

Manual Insertion of Scan-Based DFT

Counter.v

```
module counter (  
    input clk,  
    input reset,  
    input d,  
    input si,  
    input scan_en,  
    output reg q  
);  
  
always @(posedge clk or posedge reset) begin  
    if (reset)  
        q <= 1'b0;  
    else if (scan_en)  
        q <= si;  
    else  
        q <= d;  
end  
  
Endmodule
```

Counter_tb.v

```
`timescale 1ns/1ps  
  
module counter_tb;  
  
    reg clk;  
    reg reset;  
    reg scan_in;  
    reg scan_en;  
    wire scan_out;  
    wire [3:0] q;  
  
    counter_top dut (  
        .clk(clk),  
        .reset(reset),  
        .scan_in(scan_in),  
        .scan_en(scan_en),  
        .scan_out(scan_out),  
        .q(q)  
    );  
endmodule
```

```
);

// Clock generation
always #5 clk = ~clk;

initial begin
    $dumpfile("scan.vcd");
    $dumpvars(0, counter_tb);

    clk = 0;
    reset = 1;
    scan_en = 1;
    scan_in = 0;

    #20 reset = 0;

    // SHIFT IN 1011
    scan_in = 1; #10;
    scan_in = 1; #10;
    scan_in = 0; #10;
    scan_in = 1; #10;

    // CAPTURE
    scan_en = 0;
    #20;

    // SHIFT OUT
    scan_en = 1;
    #40;

    $finish;
end

endmodule
```

Counter_top.v

```
module counter_top (
    input clk,
    input reset,
    input scan_in,
    input scan_en,
    output scan_out,
    output [3:0] q
```

```
);
```

```
wire [3:0] d;  
assign d = q + 1'b1;
```

```
// Scan chain stitching  
counter ff0 (clk, reset, d[0], scan_in, scan_en, q[0]);  
counter ff1 (clk, reset, d[1], q[0], scan_en, q[1]);  
counter ff2 (clk, reset, d[2], q[1], scan_en, q[2]);  
counter ff3 (clk, reset, d[3], q[2], scan_en, q[3]);
```

```
assign scan_out = q[3];
```

```
endmodule
```

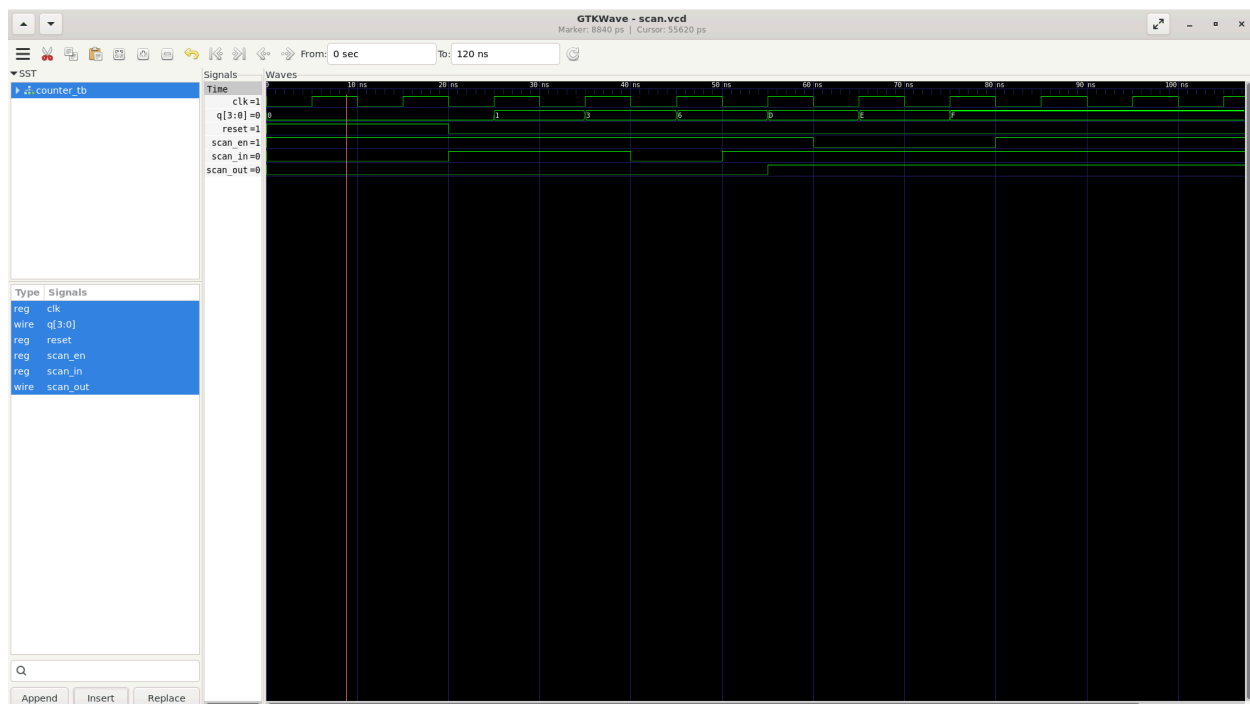
Commands Run

```
iverilog counter.v counter_top.v counter_tb.v
```

```
vvp a.out
```

```
gtkwave scan.vcd
```

GTK WAVEFORM



Open Yosys in terminal

```
read_verilog counter.v counter_top.v
hierarchy -check -top counter_top
proc
opt
techmap
synth -top counter_top
write_verilog counter_synth.v
write_blif synthesized.blif
```

[OUTPUT FILES](#)